AI COMPUTING CONTINUUM

From AI PC to AI Infrastructure: Platform Contracts for Scalable CPU–Accelerator Systems

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OCP PLATINUM Member

AICC & Arm system architecture

AICC frames the distributed AI fabric

Arm provides the repeatable system building blocks inside the fabric

Scale-Across

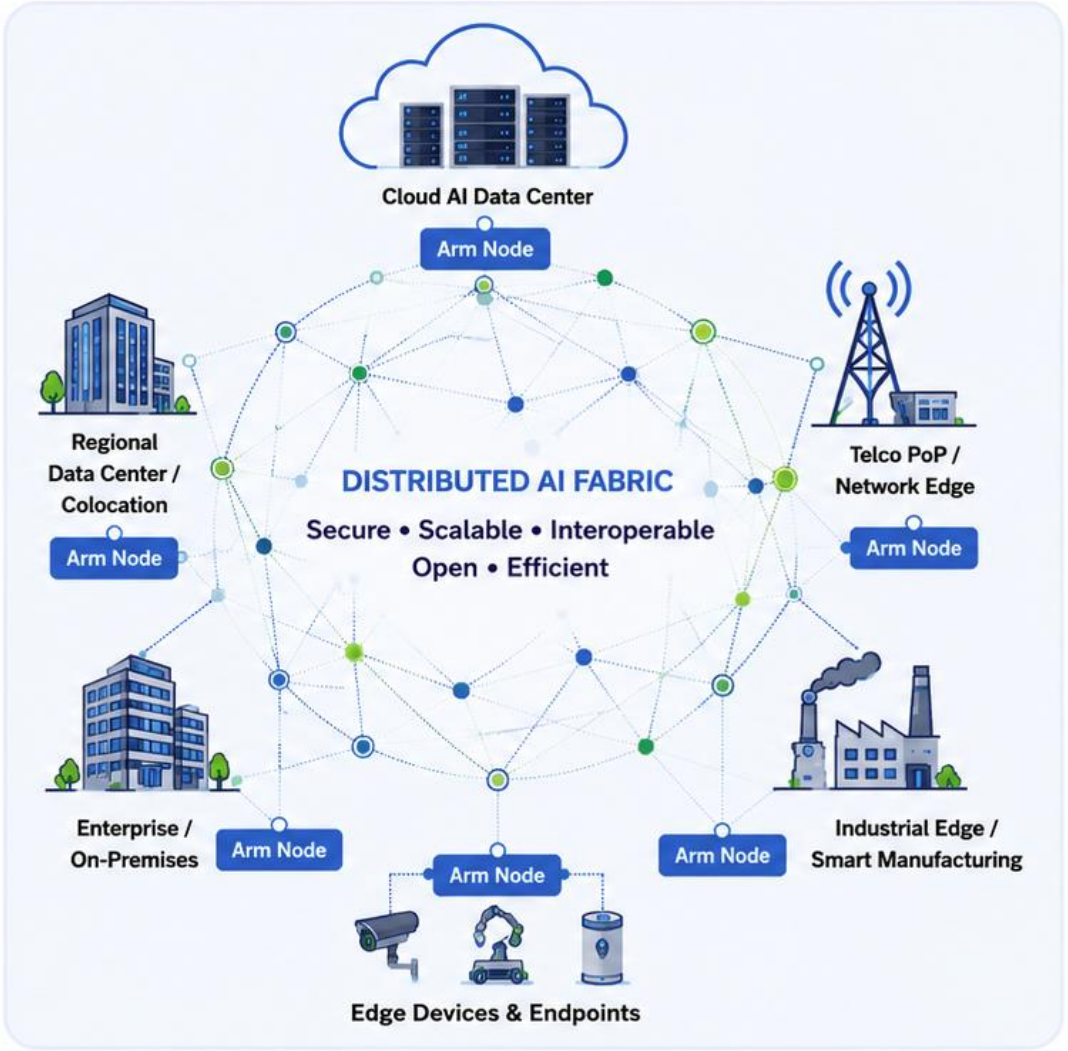
Expand AI capacity across sites

Sync-Across

Keep data, state, and operations aligned

Reach-Across

Bring AI to edge and enterprise environments



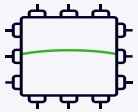
The AI infrastructure question has changed

The original scale story is still valid, but it is no longer the whole story.

YESTERDAY **Scale inside the data center**



Bigger AI clusters



More GPUs / accelerators



Larger data centers

The foundation of AI remains strong. Scaling inside data centers is still essential.

TOMORROW **Scale beyond the data center**



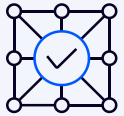
AI moves to where data and operations happen



More deployment site types



More platform diversity and deployment constraints



Repeatable Arm node foundation

AI infrastructure must be deployable across many different environments with a consistent and repeatable foundation.

AICC defines the macro-architecture for distributed AI infrastructure;
Arm system architecture provides the repeatable node foundation that can be deployed inside it.

Why node-level system architecture matters

Arm system architecture provides the foundation for distributed AI infrastructure

AICC MODEL	INFRASTRUCTURE QUESTION	ARM SYSTEM ARCHITECTURE IMPLICATION
Scale-Across	How do we efficiently expand AI capability across sites?	Use repeatable compute nodes with balanced CPU, accelerator, memory, and I/O design to scale capacity across sites.
Sync-Across	How do we keep data, state, and control aligned?	Provide coherent local data movement, telemetry, and control hooks that orchestration and software can rely on to maintain consistency across sites.
Reach-Across	How do enterprises, telco, and edge sites access AI in real time?	Deliver efficient, secure, and manageable nodes that operate reliably under diverse power, thermal, and latency constraints at the edge and at scale.

From Requirements to Deployable Nodes

AICC Requirement
What the infrastructure must enable

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System Architecture Implementation
How Arm system architecture implements the requirement

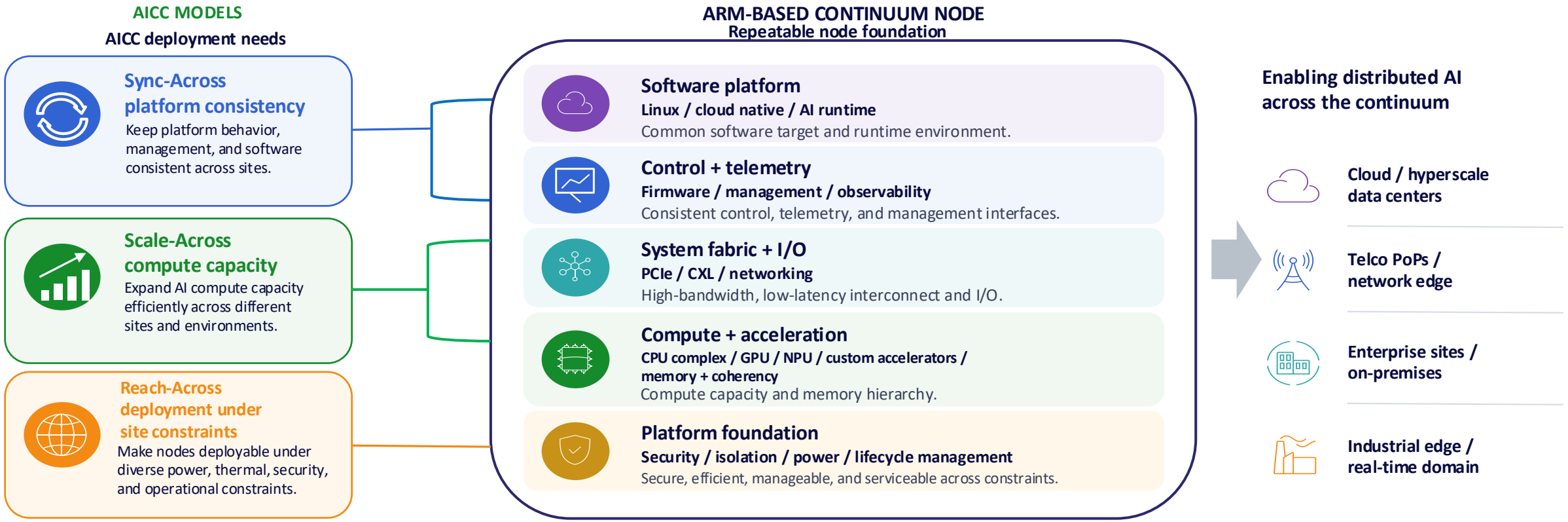
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Arm-Based Continuum Node
Deployable, repeatable node inside the distributed fabric

Arm is the node-level system substrate. It provides the **building blocks** that orchestration, software, and networking can leverage —
not a substitute for cross-site orchestration or network architecture.

Arm system architecture as the continuum node

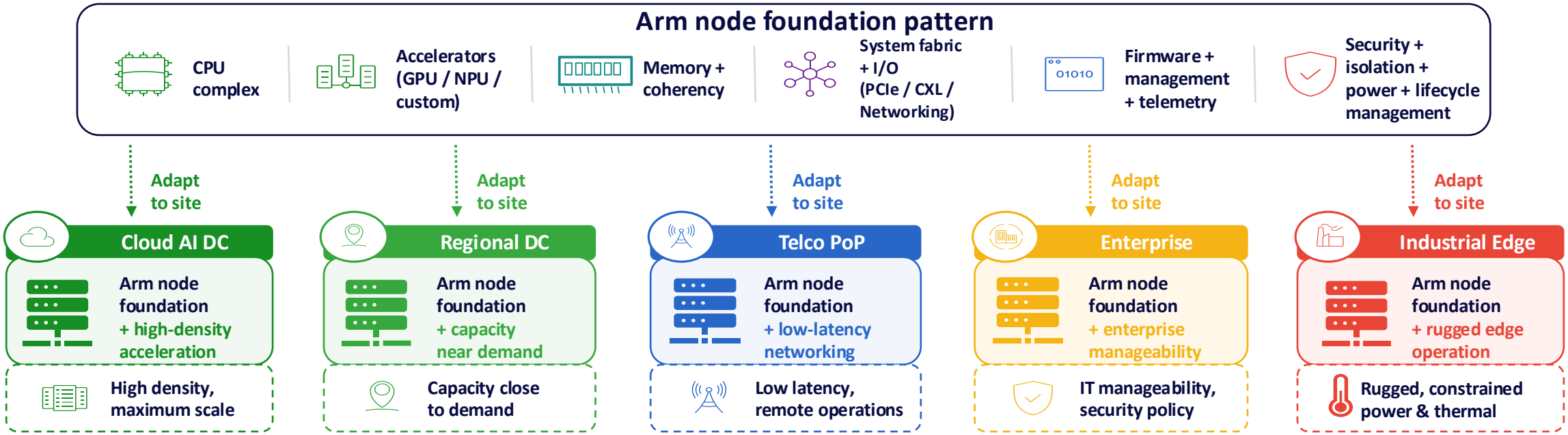
The node-level platform foundation for the AICC vision



AICC deployment needs map to different layers of the Arm node foundation: compute + acceleration and system fabric + I/O for Scale-Across, control + telemetry and software platform for Sync-Across, and platform foundation for Reach-Across.

Deploy a consistent Arm node pattern across different sites

Scale-Across means repeatable deployment across cloud, regional, telco, enterprise, and edge environments.



Architecture implications

- Common building blocks reduce per-site design variance
- Consistent CPU, accelerator, memory, and I/O patterns simplify software and operations
- Efficiency matters where power and cooling headroom are limited

A consistent Arm node pattern reduces per-site customization while allowing site-specific implementation choices.

Reach-Across Example: Enterprise-to-Edge AI Deployment

AI value appears when enterprise systems and edge operations work together on a common Arm node foundation.

Enterprise Node

(Data Center / Campus / Enterprise Site)

1. ENTERPRISE ROLE

Orchestration / policy / data services

- AI orchestration & analytics
- Data services & storage
- Policy & security
- Fleet & lifecycle management

2. USE-CASE-SPECIFIC COMPONENTS

Enterprise software, storage, networking, policy integration, applications

- Enterprise applications
- Networking & connectivity
- Storage & databases
- Integration & APIs

3. COMMON ARM NODE FOUNDATION

Compute / I/O / software / firmware / security / manageability / lifecycle

- Compute (CPU / NPU / GPU)
- Memory
- I/O & connectivity
- Security
- Manageability & lifecycle

Secure connection

- Low latency
- Reliable
- Resilient operations

Edge Node

(Factory / Hospital / Transportation / Retail / City / Telco PoP)

1. EDGE ROLE

Local inference / sensor I/O / real-time operation

- Vision AI inference
- Sensor I/O & control
- Real-time operation
- Automation & local control

2. USE-CASE-SPECIFIC COMPONENTS

Sensors, accelerators, local control, ruggedization, site-specific software & firmware

- Cameras & sensors
- Accelerators (DPU / NPU / VPU)
- Rugged enclosure & power
- Local control firmware

3. COMMON ARM NODE FOUNDATION

Compute / I/O / software / firmware / security / manageability / lifecycle

- Compute
- Memory
- I/O & connectivity
- Security
- Manageability & lifecycle

Operational Outcomes

- Quality inspection
- Safety & compliance
- Process automation & optimization
- Predictive maintenance & reliability
- Lower cost & energy efficiency

Reach-Across

Reach-Across is not only about connectivity.

It is about making enterprise and edge AI nodes deployable on a common foundation while allowing use-case-specific hardware, firmware, and software differentiation.

Enterprise node
Orchestration, policy, data services, and lifecycle management.

Edge node
Local inference, sensor I/O, and real-time operations.

Common foundation
Compute, I/O, software, firmware, security, manageability, and lifecycle.

Use-case-specific components
Hardware, software, and firmware adaptations for unique site needs and workloads.

Reach-Across connects enterprise intent with edge execution: using a common Arm foundation that preserves partner differentiation for specific use cases.

AICC and Arm system architecture

1

AICC challenge

AI infrastructure is moving beyond bigger clusters toward distributed cloud-to-edge deployment.

Distributed | **Edge-aware** | **Manageable** | **Scalable**



2

System architecture requirements

Scale-Across, Sync-Across, and Reach-Across create common node-level platform requirements.

Scale-Across
Scale efficiently across sites

Sync-Across
Maintain consistency across nodes

Reach-Across
Operate across locations

3

Arm system architecture

Arm system architecture defines a repeatable foundation for Arm-based nodes across compute, memory, I/O, security, firmware, and manageability.

Arm System Architecture Foundation
From specifications to deployable assurance

Hardware foundation	BSA / SBSA
Boot + firmware	BBR / SBBR
Security + manageability	BBSR / SBMR
Deployment assurance	SystemReady / ACS

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Deployment ecosystem

OCP provides the collaboration forum for modular AI infrastructure; Arm anchors the node-level system foundation; and SiPs, ODMs, OEMs, OSVs, CSPs, and software partners turn it into deployable systems.

OCP
(Open Collaboration) Collaboration forum & specifications

+

arm
Node-level system foundation

+

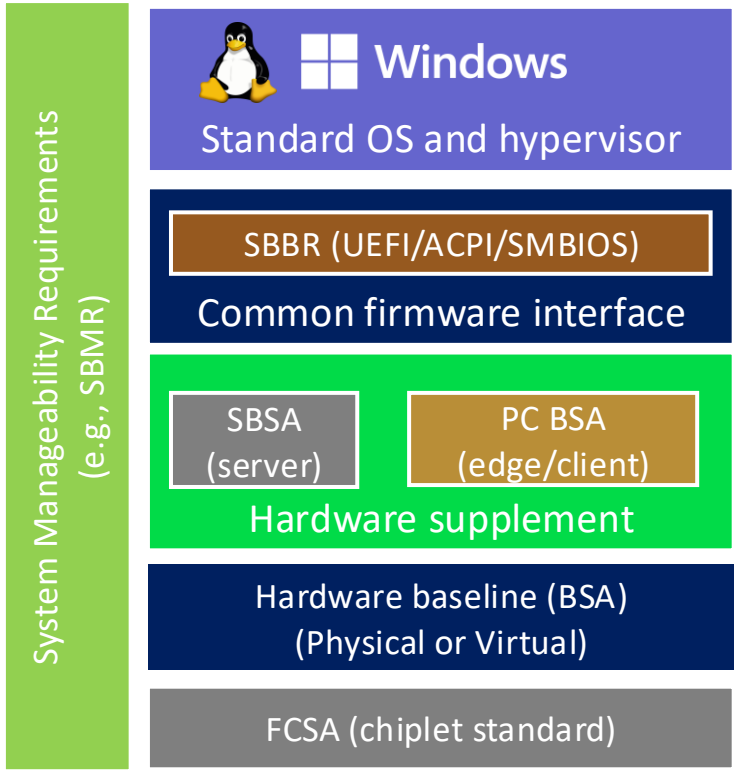
Ecosystem
(SiP/ODM/OEM/OSV/CSP/ISV) Deployable systems & solutions

AICC defines the deployment challenge.

Arm system architecture provides the common node-level foundation.

OCP, Arm, and the **ecosystem** make it deployable at scale.

Arm system architecture



CONTINUE THE DISCUSSION

SBSA: Advancing Arm Server System Architecture in the OCP Ecosystem

Today, 4:45 - 5:00 PM

SJCC - Concourse Level - Room 212

Final session in the Server track

Dong Wei and Samer El-Haj-Mahmoud

An update on the OCP SBSA workstream and the next specification release.

Arm and OCP Foundation have established SBSA and SBMR SPWs under the OCP server and Hardware Management projects, enabling the ecosystem to collaborate on future system architecture requirements.

Architecture Compliance Suite (ACS) and SystemReady™ / OCP Accepted™ help turn those requirements into testable deployment assurance.

arm SystemReady

OCP ACCEPTED™

Call to Action

Arm System Architecture in the AICC, from Cloud to Edge

Let`s Build an Open, Deployable AI Infrastructure Together

☑ **OCP defines the collaboration** ☑ **Arm provides the platform foundation** ☑ **The ecosystem delivers the solutions**

Collaborate in OCP

- Join the new SBSA & SBMR SPWs
- Contribute platform requirements
- Share deployment experience

Build on Open Architecture

- Develop repeatable Arm-based AI nodes
- Advance firmware, security & Manageability
- Enable software portability from cloud to edge

Validate & Deploy

- Adopt ACS and SystemReady™ / OCP Accepted™
- Reduce integration effort
- Accelerate ecosystem adoption

Thank You!



Demonstration



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http://www.arm.com

